

column vector. From Page 25, just below the example 1, it's mentioned that any vector $\begin{bmatrix} a \\ b \end{bmatrix}$ can be written in the form (a, b) . $\therefore$ The vector $\begin{bmatrix} -4 \\ 3 \end{bmatrix}$ can be written as $(-4, 3)$. This is not equivalent to the 1×2 row matrix $[-4 \ 3]$. $\therefore$ The statement is False $\square$

b) The points in the plane corresponding to $\begin{bmatrix} -2 \\ 5 \end{bmatrix}$ and $\begin{bmatrix} -5 \\ 2 \end{bmatrix}$ lie on a line through the origin.

Solution: The question is a bit ambiguous. Does it mean different individual lines or the same line?

c) An example of a linear combination of vectors v_1 and v_2 is the vector $\frac{1}{2}v_1$.

Solution: From Page 28, a vector y is the linear combination of vectors v_1 and v_2 if there exists $c_1, c_2 \in \mathbb{R}$ such that $c_1v_1 + c_2v_2 = y$.

$\frac{1}{2}v_1 = \frac{1}{2}v_1 + 0v_2$. We know, $\frac{1}{2}$ and 0 are real numbers.

By defn, we can say that $\frac{1}{2}v_1$ vector is a linear combination of the vectors v_1 and v_2 . $\therefore$ The statement is True. $\square$

d) The solution set of the linear system whose augmented matrix is $\begin{bmatrix} a_1 & a_2 & a_3 & b \end{bmatrix}$ is the same as the solution set of the equation $x_1a_1 + x_2a_2 + x_3a_3 = b$.

Soln: On Page 30 inside the topmost grey box it is mentioned that to a vector equation $x_1a_1 + x_2a_2 + \dots + x_n a_n = b$ has the same solution set as the linear system whose augmented matrix is $\begin{bmatrix} a_1 & a_2 & \dots & a_n & b \end{bmatrix}$. $\therefore$ The given statement is True since, it's the particular representation of $n=3$ from general n .

e) The set $\text{Span}\{u, v\}$ is always visualized as a plane through the origin.

Soln: On Page 30 under the section "A Geometric Description of $\text{Span}\{u, v\}$ and $\text{Span}\{u, v, w\}$ " it is mentioned that If u and v are